PROBLEM 35

String with Two Particles on a Cylinder

Original statement · Problems.pdf, p. 23

Multi-Particle Cylinder Contact

Two heavy particles P and Q of mass m each are connected by an inextensible massless string of constant length L. Part of this string rests on top of a cylinder of radius R and center at O. The string can slide over the cylinder without friction, and remains tense at all times. Furthermore, P and Q are also connected by a spring of zero natural length and stiffness k. We define the inertial reference frame S_0. We denote as A and B the points of tangency of the string with the cylinder.

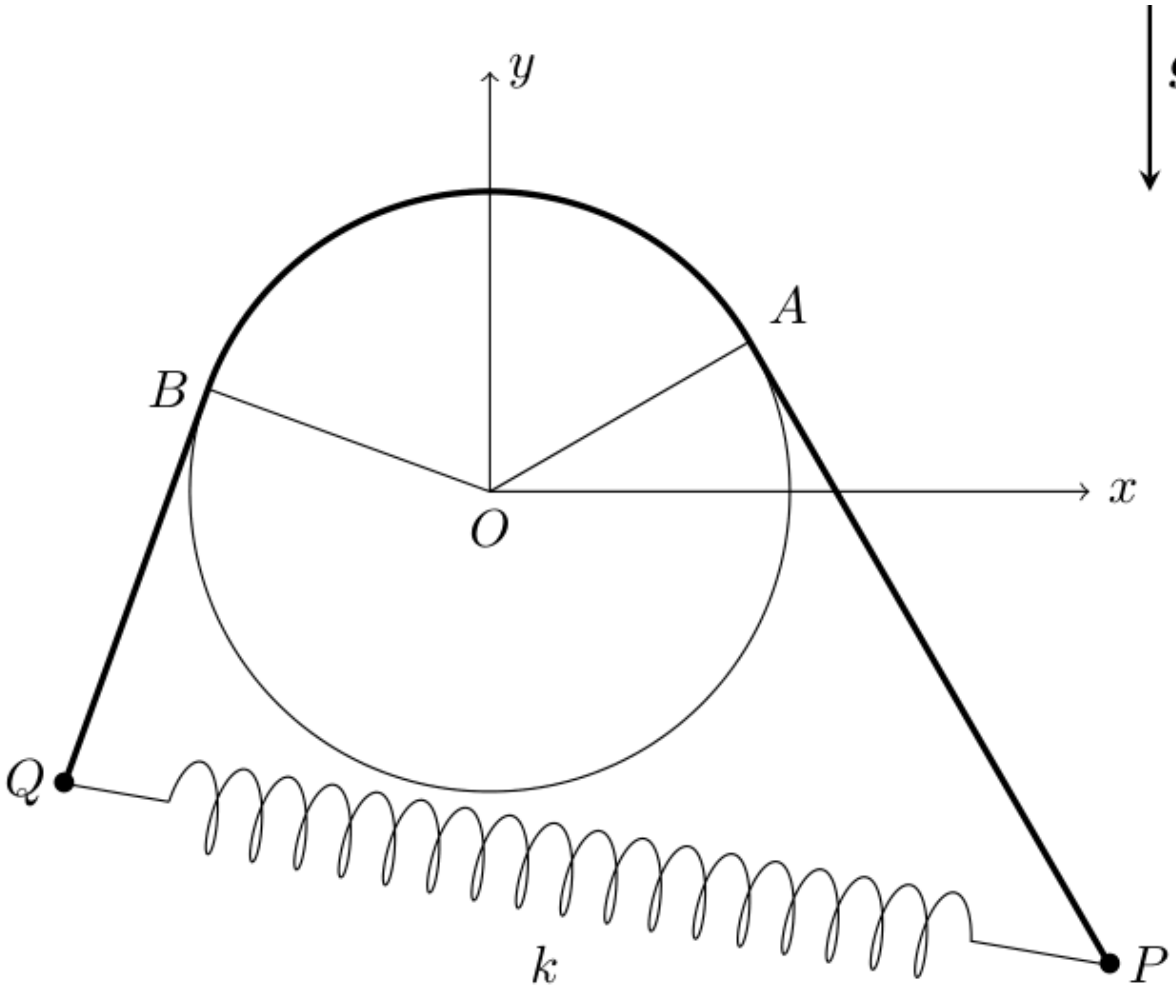


Figure: String over the cylinder of radius R centred at O with tangency points A and B, particles P and Q joined by the spring k (Problems.pdf, p. 23).

(a) How many degrees of freedom does this system have? Define your own generalized coordinates to parametrize the configuration of the system. You will use them in the rest of the problem.

(b) Write down the general expression of the position, velocity, and acceleration vectors of P and Q relative to the S_0 reference frame as a function of your chosen coordinates and their time derivatives.

(c) Write down the expression of all forces acting on P and Q.

(d) Discuss whether the mechanical energy of P is conserved; if so, write down the expression of the corresponding potential energy.

(e) Consider now that we restrict the motion of the system to symmetric configurations about the Oy axis only. What is the number of degrees of freedom now? Find the mathematical relations that exist between your previously-defined coordinates in this case.

(f) Find the approximate value of your coordinates when the system is in equilibrium and the configuration is symmetric about the Oy axis. To do so, assume that the deviation from the vertical of the AP and BQ segments is small, and linearize the equations of motion. Find the angular frequency of the oscillations around this equilibrium configuration.

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** two particles P and Q of mass m joined by an inextensible, massless string of length L that rests on a fixed cylinder of radius R and centre O, and by a spring of zero natural length and stiffness k. The motion is planar, in the plane Oxy with Oy upward.
- **Hypotheses:** the cylinder is smooth, the string is always tense and touches the cylinder along the arc between the tangency points A (right) and B (left), the spring does not touch the cylinder, and the inertial frame $S_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ is fixed to the cylinder.
- **Parameters:** m [kg], L [m], R [m], k [N/m], g [m/s^2]; $\ell_0 = (L - \pi R)/2$ [m] is defined below.

PHASE 2

Reference Frames, Vector Bases & Coordinate Geometry

Tangency point A is at the polar angle α from Ox and B at the polar angle β from Ox. Two orthonormal bases follow the contact points:

$$\mathbf{n}_A = \cos\alpha\,\mathbf{i} + \sin\alpha\,\mathbf{j}, \quad \mathbf{t}_A = \sin\alpha\,\mathbf{i} - \cos\alpha\,\mathbf{j}, \qquad \mathbf{n}_B = \cos\beta\,\mathbf{i} + \sin\beta\,\mathbf{j}, \quad \mathbf{t}_B = -\sin\beta\,\mathbf{i} + \cos\beta\,\mathbf{j}$$

$\mathbf{n}$ is the outward normal of the cylinder and $\mathbf{t}$ the unit vector along the straight string, pointing from the tangency point to the particle (clockwise at A, counter-clockwise at B). Change of basis from $\{\mathbf{i},\mathbf{j},\mathbf{k}\}$ (columns are the components of $\mathbf{t}_A,\mathbf{n}_A,\mathbf{k}$ and of $\mathbf{n}_B,\mathbf{t}_B,\mathbf{k}$):

$$[{}_0R_A]=\begin{pmatrix}\sin\alpha & \cos\alpha & 0\\ -\cos\alpha & \sin\alpha & 0\\ 0 & 0 & 1\end{pmatrix},\qquad [{}_0R_B]=\begin{pmatrix}\cos\beta & -\sin\beta & 0\\ \sin\beta & \cos\beta & 0\\ 0 & 0 & 1\end{pmatrix}$$

$$\det[{}_0R_A]=\sin^2\alpha+\cos^2\alpha=1,\quad \det[{}_0R_B]=\cos^2\beta+\sin^2\beta=1,\quad [{}_0R_A][{}_0R_A]^T=[{}_0R_B][{}_0R_B]^T=I$$

Time derivatives (each basis turns about $\mathbf{k}$ at the rate of its angle):

$$\dot{\mathbf{t}}_A=\dot\alpha\,\mathbf{n}_A,\quad \dot{\mathbf{n}}_A=-\dot\alpha\,\mathbf{t}_A,\qquad \dot{\mathbf{n}}_B=\dot\beta\,\mathbf{t}_B,\quad \dot{\mathbf{t}}_B=-\dot\beta\,\mathbf{n}_B$$

PHASE 3

Step-by-Step Mathematical Deduction

(a) Degrees of freedom and coordinates

Pedagogical Justification: Two free particles in the plane have 4 degrees of freedom. The inextensible string removes one, so the system has 3.

The straight segments AP and BQ have lengths ξ_P and ξ_Q, and the string wraps the arc AB of length $R(\beta-\alpha)$ on the cylinder. The length constraint is

$$\xi_P+\xi_Q+R(\beta-\alpha)=L$$

Generalized coordinates: (α,β,ξ_P), with $\xi_Q=L-\xi_P-R(\beta-\alpha)$ (the quadruple $(\alpha,\beta,\xi_P,\xi_Q)$ is used below with this one relation).

(b) Position, velocity and acceleration

Positions: $\mathbf{r}_P = R\,\mathbf{n}_A + \xi_P\,\mathbf{t}_A$ and $\mathbf{r}_Q = R\,\mathbf{n}_B + \xi_Q\,\mathbf{t}_B$.

Velocity of P (product rule on each term, using $\dot{\mathbf{n}}_A = -\dot\alpha\,\mathbf{t}_A$ and $\dot{\mathbf{t}}_A = \dot\alpha\,\mathbf{n}_A$):

$$\mathbf{v}_P = -R\dot\alpha\,\mathbf{t}_A + \dot\xi_P\,\mathbf{t}_A + \xi_P\dot\alpha\,\mathbf{n}_A = (\dot\xi_P - R\dot\alpha)\,\mathbf{t}_A + \xi_P\dot\alpha\,\mathbf{n}_A$$

Differentiate again, again with the product rule:

$$\mathbf{a}_P = (\ddot\xi_P - R\ddot\alpha)\,\mathbf{t}_A + (\dot\xi_P - R\dot\alpha)\dot\alpha\,\mathbf{n}_A + (\dot\xi_P\dot\alpha + \xi_P\ddot\alpha)\,\mathbf{n}_A - \xi_P\dot\alpha^2\,\mathbf{t}_A$$

$$\mathbf{a}_P = \left(\ddot\xi_P - R\ddot\alpha - \xi_P\dot\alpha^2\right)\mathbf{t}_A + \left(\xi_P\ddot\alpha + 2\dot\xi_P\dot\alpha - R\dot\alpha^2\right)\mathbf{n}_A$$

For Q, with $\dot{\mathbf{n}}_B = \dot\beta\,\mathbf{t}_B$ and $\dot{\mathbf{t}}_B = -\dot\beta\,\mathbf{n}_B$:

$$\mathbf{v}_Q = (\dot\xi_Q + R\dot\beta)\,\mathbf{t}_B - \xi_Q\dot\beta\,\mathbf{n}_B$$

$$\mathbf{a}_Q = \left(\ddot\xi_Q + R\ddot\beta - \xi_Q\dot\beta^2\right)\mathbf{t}_B - \left(\xi_Q\ddot\beta + 2\dot\xi_Q\dot\beta + R\dot\beta^2\right)\mathbf{n}_B$$

These four components were checked symbolically with sympy (zero residual).

(c) Forces

- Weights: $-mg\,\mathbf{j}$ on each particle.
- String tension $T \geq 0$ (the same on both sides because the cylinder is smooth and the string massless): $-T\,\mathbf{t}_A$ on P and $-T\,\mathbf{t}_B$ on Q, directed towards the tangency points. The cylinder acts on the string, not on the particles.
- Spring (zero natural length): $\mathbf{F}_{s,P} = k(\mathbf{r}_Q - \mathbf{r}_P)$ on P and $\mathbf{F}_{s,Q} = -\mathbf{F}_{s,P}$ on Q.

Projections of the weight and of the spring force on the bases ($\mathbf{j}\cdot\mathbf{t}_A = -\cos\alpha, \mathbf{j}\cdot\mathbf{n}_A = \sin\alpha, \mathbf{j}\cdot\mathbf{t}_B = \cos\beta, \mathbf{j}\cdot\mathbf{n}_B = \sin\beta$) give the four Newton equations. Writing $\mathbf{d} = \mathbf{r}_Q - \mathbf{r}_P$:

$$m\left(\ddot{\xi}_P - R\ddot{\alpha} - \xi_P\dot{\alpha}^2\right) = -T + mg\cos\alpha + k\,\mathbf{d}\cdot\mathbf{t}_A$$

$$m\left(\xi_P\ddot{\alpha} + 2\dot{\xi}_P\dot{\alpha} - R\dot{\alpha}^2\right) = -mg\sin\alpha + k\,\mathbf{d}\cdot\mathbf{n}_A$$

$$m\left(\ddot{\xi}_Q + R\ddot{\beta} - \xi_Q\dot{\beta}^2\right) = -T - mg\cos\beta - k\,\mathbf{d}\cdot\mathbf{t}_B$$

$$-m\left(\xi_Q\ddot{\beta} + 2\dot{\xi}_Q\dot{\beta} + R\dot{\beta}^2\right) = -mg\sin\beta - k\,\mathbf{d}\cdot\mathbf{n}_B$$

Together with the length constraint these are 5 equations for the 5 unknowns $\alpha, \beta, \xi_P, \xi_Q, T$. Subtracting the first and third equations removes T.

(d) Mechanical energy

> **Pedagogical Justification:** The mechanical energy of P alone is not conserved: the tension and the spring do work on it, and that work is exchanged with Q. For the pair $P+Q$ the only forces that could dissipate energy are the tensions, so their total power is computed first.

Using $\mathbf{t}_A\cdot\mathbf{v}_P = \dot{\xi}_P - R\dot{\alpha}$ and $\mathbf{t}_B\cdot\mathbf{v}_Q = \dot{\xi}_Q + R\dot{\beta}$:

$$\dot{W}_T = -T\,\mathbf{t}_A\cdot\mathbf{v}_P - T\,\mathbf{t}_B\cdot\mathbf{v}_Q = -T\,\frac{d}{dt}\left[\xi_P + \xi_Q + R(\beta-\alpha)\right] = -T\,\dot{L} = 0$$

The tensions do zero net work. Gravity and the spring are conservative, so the total mechanical energy of the system is conserved, with the potential energy

$$V = mg\,(y_P + y_Q) + \tfrac{1}{2}k\,|\mathbf{r}_Q - \mathbf{r}_P|^2, \qquad E = \tfrac{1}{2}m\,(v_P^2 + v_Q^2) + V = \text{constant}$$

(e) Symmetric configurations about Oy

If the motion keeps the mirror symmetry about Oy, then $\beta = \pi - \alpha$ and $\xi_P = \xi_Q = \xi$. The length constraint becomes $2\xi + R(\pi - 2\alpha) = L$, that is

$$\xi(\alpha) = \ell_0 + R\alpha, \qquad \ell_0 = \frac{L - \pi R}{2}$$

The system has 1 degree of freedom, the angle α. The relations between the coordinates are $\beta = \pi - \alpha$, $\xi_P = \xi_Q = \ell_0 + R\alpha$ and $\dot{\xi} = R\dot{\alpha}$.

(f) Equilibrium and small oscillations

Coordinates of P in the symmetric case, with $\xi = \ell_0 + R\alpha$:

$$x_P = R\cos\alpha + \xi\sin\alpha, \qquad y_P = R\sin\alpha - \xi\cos\alpha$$

Derivatives with respect to α (note $\xi' = R$):

$$x_P' = -R\sin\alpha + R\sin\alpha + \xi\cos\alpha = \xi\cos\alpha, \qquad y_P' = R\cos\alpha - R\cos\alpha + \xi\sin\alpha = \xi\sin\alpha$$

Hence $v_P = \xi\dot{\alpha}$. By symmetry $x_Q = -x_P$, $y_Q = y_P$, $|\mathbf{r}_Q - \mathbf{r}_P| = 2x_P$, and

$$T_{kin} = 2\cdot\tfrac{1}{2}m\,\xi^2\dot{\alpha}^2 = m\,\xi^2\dot{\alpha}^2, \qquad V(\alpha) = 2mg\,y_P + \tfrac{1}{2}k\,(2x_P)^2 = 2mg\,y_P + 2k\,x_P^2$$

$$V'(\alpha) = 2mg\,\xi\sin\alpha + 4k\,x_P\,\xi\cos\alpha = 2\xi\,f(\alpha), \qquad f(\alpha) = mg\sin\alpha + 2k\cos\alpha\,(R\cos\alpha + \xi\sin\alpha)$$

Pedagogical Justification: Equilibrium requires $V'(\alpha_0)=0$, that is $f(\alpha_0)=0$ (since $\xi>0$). Physically, the horizontal pull $2kx_P$ of the spring must be balanced by the horizontal component of the tension, which needs $\alpha_0<0$: the segments AP and BQ are slightly inclined inwards.

Assuming the deviation of AP and BQ from the vertical is small, $|\alpha_0|\ll 1$, keep first-order terms ($\sin\alpha\approx\alpha$, $\cos\alpha\approx 1$, $\xi\approx\ell_0$):

$$mg\,\alpha_0 + 2k\,(R+\ell_0\alpha_0)=0 \;\Longrightarrow\; \alpha_0\approx -\frac{2kR}{mg+2k\ell_0}$$

The equation of motion follows from $\frac{d}{dt}(T_{kin}+V)=0$. Differentiate with the chain rule, using $\dot\xi = R\dot\alpha$:

$$2m\xi^2\dot\alpha\ddot\alpha + 2m\xi R\,\dot\alpha^3 + 2\xi f\,\dot\alpha = 0 \;\Longrightarrow\; m\,\xi\,\ddot\alpha + mR\,\dot\alpha^2 + f(\alpha)=0$$

This is exactly the $\mathbf{n}_A$ equation of part (c) with $\dot\xi_P = R\dot\alpha$ and $\mathbf{d}\cdot\mathbf{n}_A = -2x_P\cos\alpha$. Put $\alpha=\alpha_0+\delta$ with δ small; the term $\dot\alpha^2$ is second order and $f(\alpha)\approx f'(\alpha_0)\,\delta$:

$$f'(\alpha) = mg\cos\alpha + 2k\left[\xi\cos^2\alpha - x_P\sin\alpha\right] \approx mg + 2k\,\ell_0$$

$$\ddot\delta + \frac{mg+2k\ell_0}{m\,\ell_0}\,\delta = 0 \;\Longrightarrow\; \omega^2 = \frac{g}{\ell_0}+\frac{2k}{m}$$

$$\boxed{\omega=\sqrt{\frac{g}{\ell_0}+\frac{2k}{m}},\qquad \ell_0=\frac{L-\pi R}{2}}$$

PHASE 4

Physical Interpretation & Dimensional Verification

- **Limit $k \to 0$:** $\omega = \sqrt{g/\ell_0}$, the pendulum of length ℓ_0, the free length of each hanging segment. The string unwinds from the cylinder, but at the vertical the instantaneous centre of rotation of P is the tangency point A, at distance $\xi_0 \approx \ell_0$.
- **Limit $g \to 0$:** $\omega = \sqrt{2k/m}$, the frequency of two particles of mass m joined by a spring k (reduced mass $m/2$ and stiffness k).
- **Dimensional check:** $[g/\ell_0] = \mathrm{s}^{-2}$, $[2k/m] = \mathrm{N/m/kg} = \mathrm{s}^{-2}$, so $[\omega] = \mathrm{s}^{-1}$; $[\alpha_0] = [kR/(mg + 2k\ell_0)] = \mathrm{N/N} = 1$.
- **Numerical check ($R = 1$ **m**, $\ell_0 = 2$ **m**, $m = 1$ **kg**, $k = 0.3$ **N/m**, $g = 9.81$ **m/s^2**):** the exact root of $f(\alpha_0) = 0$ is $\alpha_0 = -0.05453$ rad against -0.05450 rad from the linear formula, and the exact $\omega^2 = f'(\alpha_0)/(m\,\xi(\alpha_0)) = 5.648\ \mathrm{s}^{-2}$ against $g/\ell_0 + 2k/m = 5.505\ \mathrm{s}^{-2}$; the small difference is the first-order correction $\xi(\alpha_0) = \ell_0 + R\alpha_0$ in place of ℓ_0 (using $\xi(\alpha_0)$ gives $5.642\ \mathrm{s}^{-2}$).

Source: Problems.pdf (Page 23)